CHAPTER

30

Management of Deep Carious Lesions

In operative dentistry, the major focus has always been on improving the technical aspects of cavity design and choice of restorative materials. However, management of progression of caries along with its pathologic features is important for the success of any restorative treatment.

Vital pulp is essential for routine restorative procedures; therefore, every precaution should be taken to preserve the vitality of tooth.

As the caries progress from initial enamel/cemental caries towards pulp via dentin, the tooth tissues react as per body defence mechanism. The healing process prevails when injurious agent (caries) is removed, followed by initiating therapeutic measures.

The changes seen in dentin–pulp complex with advancing caries are:

a. ***Advancing carious lesion in enamel***

- First cellular change subjacent to active progressing enamel lesions is the reduction of odontoblasts around the predentin region.
- Size of odontoblasts also decreases.
- Pulpal cells proliferate into the cell-free zone.
- Demineralization in enamel approaching dentino-enamel junction; may lead to dentin tubular hypermineralization.

b. ***Advancing carious lesion in dentin***

- Demineralization of dentin is initiated, as the enamel lesion comes in contact with the dentino-enamel junction. This dentin demineralization is a reaction of bacterial byproducts passing through enamel prisms.
- The advancing front of dentinal caries follows the direction of dentinal tubules, this being the primary route for dissolution of the hard dentinal tissue. Consequently, the zone of dentin demineralization subjacent to non-cavitated lesions narrows, as it progresses toward the pulp.
- The initial dentin demineralization takes place without the presence of bacteria in the dentin. However, when the enamel crumbles, bacterial invasion of the demineralized enamel occurs.

Effective Depth (Remaining Dentin Thickness)

The effective depth is the area of minimum thickness of sound dentin separating the pulpal tissue from the carious lesion.

If remaining dentin thickness is more than 1.0 mm, no significant disturbance in the pulp is seen. 0.25 to 0.5 mm remaining dentin leads to maximal reparative dentin formation and remineralization of demineralized dentin; whereas, 0.25 mm or less remaining dentin provokes bacterial growth in pulp. Reparative dentin formation and remineralization is greatly decreased.

Caries may progress as an active [acute (fast progressing) lesion, or chronic, (slow progressing)] lesion or may attain quiescence as arrested caries. The differences between acute/fast progressing and chronic/slow progressing carious lesion are summarized in Table 30.1.

ARRESTED CARIES

The conditions under which an active lesion gets arrested are still not clear. An arrested dentinal lesion differs from an active lesion by its darker pigmentation, absence of visible bacteria within the tubule and impermeability to dyes and isotopes.

In arrested lesions, the bacterial bodies are coalesced into homogenous masses, degenerated and disintegrated. Another characteristic is the presence of deep pigmentation. The degenerated bacteria or the degradation products of their proteins and nucleic acids are the possible sources of pigmentation.

Following layers are identified in an arrested lesion:

1. A narrow surface layer, brown in color and leathery in consistency.
2. The widest zone, dark brown in color and hard in consistency.
3. Sclerotic layer, white in color and often harder than normal dentin.

Table 30.1: Differences between acute and chronic carious lesions

Acute caries/fast progressing	*Chronic caries/slow progressing*
• Dentin is yellowish, light brown in color	• Dentin is dark brown in color
• Hardness of dentin decreases considerably	• Decrease in hardness relatively less
• Proliferation of cells takes place from the subodontoblastic cell-rich zone into cell-free zone.	• Not evident in slow progressing lesions
• Odontolbasts necrosis is faster, followed by atubular dentin/fibrodentin deposition by newly formed odontoblasts	• Tertiary dentin formed is tubular dentin at the pulpal end; as a result of reactionary dentinogenesis and by new odontoblast like cells
• Rapid demineralization, followed by tissue destruction	• Organic tissue exhibits more resistance; defensive process predominates

One of the major characteristics of an arrested caries is its high degree of mineralization. The possible ways by which the minerals get accumulated can be:

a. Remineralization of the surface layer—source can be saliva, from where calcium and phosphate are absorbed.
b. Re-precipitation of dissolved apatite—the minerals dissolved by bacterial acids in the upper demineralized layer re-precipitate to form larger crystals across the demineralized layer.
c. Intratubular calcification and obliteration of tubules underneath the lesion. Probably minerals are mediated by odontoblastic processes through blood supply.

REPAIR OF PULP–DENTIN COMPLEX

The reparative capacity of pulp–dentin complex determines the fate of advancing carious lesion. It is dependent on following factors.

A. Status of Dentin

Presence of a calcific barrier/sclerotic dentin indicates formation of reparative dentin. Generalized discolored dentin ranging from greyish-to-greyish-brown may indicate necrosis of underlying pulp.

B. Status of Pulp

1. *Pain*

Pain could serve as a tool to assess the pulpal status. The presence or absence of pain varies with different pathological conditions of pulp.

a. Spontaneous pain not initiated by thermal/chemical stimulation is indicative of pulp–dentin degeneration.
b. Pain initiated by thermal/chemical stimulation of pulp–dentin organ that disappears immediately after the removal of the stimulation is an indication of lesser degree of degenerative changes.

2. *Radiographs*

The radiographs are important tools and significantly evaluates:

a. *Proximity of the carious lesion* to the pulp chamber and root canal system.
b. *Effective depth* (thickness of the remaining dentin).
c. *Any pulpal changes* in the form of intrapulpal and peripulpal calcification (denotes reduction of the reparative capacity of the pulp; as it reduces/prevents flow of fluids into or out of the pulp/root canal system).
d. *Periradicular status:* The thickening of the periodontal ligament with an intact laminadura, indicates increased vascularity and consequently increased reparative capacity. Discontinuity in lamina dura indicates degeneration.
e. *Size of pulp chamber compared to size of tooth:* The higher the pulp size/tooth ratio, the better will be the reparative capabilities of the pulp–dentin organ (Selective high ratio in comparison to other teeth may indicate cessation of reparative capacity).
f. *Permeability of dentin:* Sclerotic dentin and calcific barrier can be identified as radiopaque regions. Tertiary dentin appears as a localized thickening of the dentinal bridge.

3. *Pulp Testing*

a. *Thermal pulp testing:* Application of heat or cold stimulates the nerves present in the pulp; helps in arriving at a primitive diagnosis regarding the condition of the pulp.
b. *Electric pulp testing:* The electric excitation of the neural elements by subjecting the tooth to an increased degree of electric current may show positive response (vitality of pulp) or no response (pulp necrosis).

C. Extent of Pulp Exposure

a. *Size of pulp exposure*: Size of pulp exposure in relation to pulp chamber/root canal system is a

major factor determining the reparability via dentinal bridging.

b. *Pinpoint exposure:* Sound dentin at periphery, no hemorrhage, vital pulp, no pulpal inflammation or mild inflammation at the exposure site, all features of successful repair.

The features suggestive of doubtful repair are:

- Exposure having infected dentin at periphery—reparability doubtful.
- Exposure with profuse hemorrhage—beyond repair
- Exposure with inflammatory fluid/pus—beyond repair.
- Exposure near the anatomical constrictions in root canals—lesser will be the reparability because of the diminished availability of nutrients.

TREATMENT MODALITIES OF DEEP CARIES LESIONS

Various treatment modalities are:

1. Indirect pulp capping/indirect pulp treatment
2. Direct pulp capping
3. Pulp curettage
4. Partial pulpotomy
5. Pulpotomy

1. Indirect Pulp Capping/Indirect Pulp Treatment

Indirect pulp capping implies treating pulps, which are near exposure; whereas, indirect pulp treatment is a technique in which an effort is made to avoid exposure during treatment of deep carious lesions without any evidence of periapical pathology. Indirect pulp capping is a two-appointment procedure; whereas, indirect pulp treatment is a single appointment procedure.

Indications

- Deep carious lesion with minimal pulpal inflammation.
- Where complete removal of caries may lead to pulp exposure.

Contraindications

- Tooth with existing pulp inflammation
- Tooth with periapical pathosis

Objectives

- Minimize pulp injury
- Promote dentin sclerosis
- Minimize postoperative sensitivity
- Arrest carious lesion
- Stimulation of reparative dentin formation
- Remineralize affected dentin

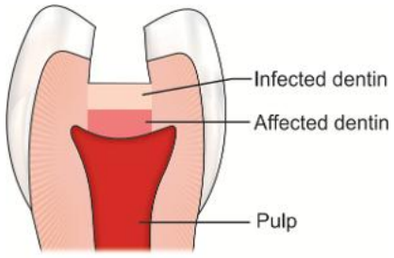


Fig. 30.1: Infected and affected dentin

Rationale

In a carious lesion, dentin decalcification precedes bacterial invasion within dentin. In an active carious lesion, two distinct layers are seen (Fig. 30.1):

i. *Infected dentin:* It is necrotic, soft dentin, not painful to stimulation and grossly infected with bacteria. Collagen is irreversibly denatured.

ii. *Affected dentin:* It is demineralized, discolored but hard dentin, painful to stimulation containing a few bacteria. It is reversibly denatured and capable of being remineralized.

When outer layer is removed, most of the bacteria are removed and the substrate on which they thrive is also removed. When cavity is sealed with a suitable material, any remaining bacteria are either killed or lie in a stage of dormancy. However, affected dentin can be left on pulpal and axial walls and never on dentinoenamel junction and cavity margins. The two layers can be differentiated with either 0.5% basic fuchsin or 1.0% acid red dyes. Acid red is preferred, since basic fuchsin has carcinogenic potential.

Various types of dyes are used to indicate the extent of progress of caries. 0.5% basic fuchsin in propylene glycol applied to dentin for 10 seconds and washing it off with water may guide an operator as to where to stop in excavating caries from cavity walls and floor.

It is the nature of collagen fibers, which differ in infected and affected dentin, that accounts for their different stainability. The affected dentin mainly have intact collagen fibers oriented for remineralization and may not stain with dye; however, infected dentin with irreparable damaged collagen, stains red.

Procedure

Procedure can be carried out either in two appointments or in one appointment.

1. Two-appointment Protocol

First sitting: Local anesthesia is administered and the tooth is isolated with a rubber dam. Cavity outline is established with appropriate instrumentation. Peripheral carious dentin/infected dentin should be

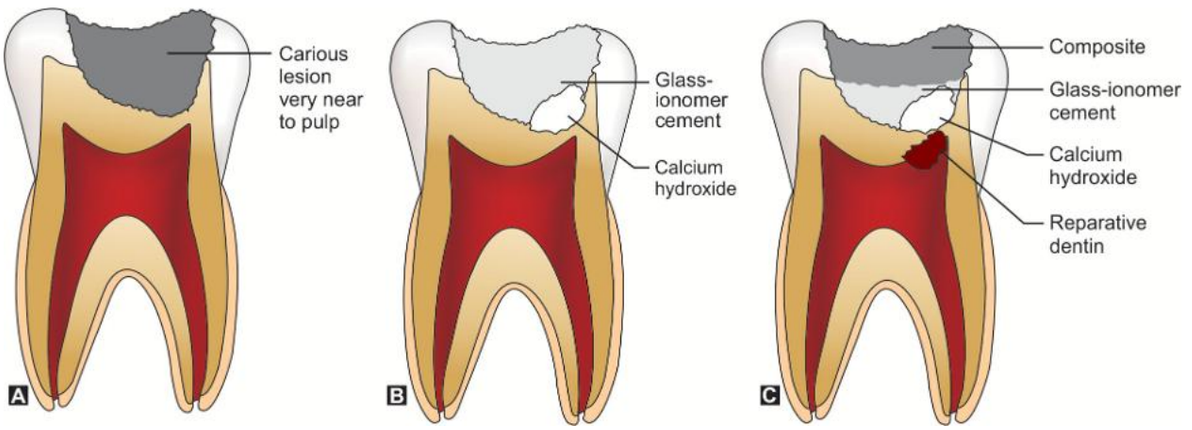


Fig. 30.2: Indirect pulp capping: **(A)** Caries approaching pulp; **(B)** Area covered with calcium hydroxide; **(C)** Formation of reparative dentin

removed with sharp spoon excavators. A few authors prefer using large round bur. Cavity should be irrigated and dried with cotton followed by placement of hard setting calcium hydroxide over the remaining affected dentin. The remainder of the cavity is filled with reinforced zinc oxide eugenol cement or glass-ionomer cement. The seal should not be disturbed for 6–8 weeks, as the carious process in the deeper layer gets arrested.

Second sitting (6–8 weeks later): In the second sitting, the temporary filling material along with calcium hydroxide dressing is removed. The color changes from deep red rose to light gray/light brown. The texture would change from spongy to hard (caries appear as dehydrated). The dehydrated and 'flaky' caries should be removed. Do not disturb predentin, which is the area around the potential exposure appearing whitish and may be soft. The cavity preparation is restored with suitable material (Fig. 30.2A to C).

The process involved in second sitting is controversial. It is established that the affected caries gets mineralized within the restoration. Re-entry should only be carried out, if the patient is asymptomatic.

With re-entry, there might be a risk of creating pulp exposure and further insult to the pulp. If pulp exposure occurs during re-entry, a more invasive vital pulp therapy techniques, such as, direct pulp capping or pulpotomy would be indicated.

2. One-appointment Protocol

The one-appointment indirect pulp treatment prefer removing caries in one appointment (excavate caries as close as possible to pulp), i.e. some caries be left to avoid an exposure. A protective liner is placed and the tooth is restored. No subsequent re-entry is carried out to remove residual caries.

Stepwise Excavation of Caries

The stepwise excavation of deep caries has shown to be successful in managing reversible pulpitis without pulpal involvement.

a. The first step is removal of the carious dentin along the dentinoenamel junction (DEJ) and excavation of only the outermost infected dentin. The remaining carious mass is covered with pulp capping material and restored with interim restorative material. The objective is to change the cariogenic environment in order to decrease the number of bacteria, and slow/arrest the caries development. It is emphasized that changing the cavity environment from an active lesion into the slowly progressing lesion, a more regular and tubular tertiary dentin formation occurs.
b. The second step is excavating the remaining caries after removal of interim material. The interval between steps is 3–6 months, allowing sufficient time for the formation of tertiary dentin. Critical to both steps of excavation is the placement of a well-sealed restoration. The technique to be followed is operator's preference.

2. Direct Pulp Capping

Direct pulp capping implies covering the exposed pulp with appropriate pulp capping material. Exposure caused by trauma or by operators' fault can be successfully treated; whereas, exposure caused by caries usually need root canal treatment. The younger the patient, the better are the chances of healing and repair.

Indications

- Size of exposure should be less than 0.5 mm
- No profuse hemorrhage or serous/purulent exudates
- In case of traumatic exposure, early treatment is indicated.

- Non-tender to percussion (a recently traumatized tooth may be reversibly tender to percussion).

Contraindications

- Instrument penetrating the pulp; infected dentin chips may settle in the pulp.
- Tooth should not be periodontally involved.
- Symptoms of irreversible pulpitis.

Ideal Requirements of Pulp Capping Material

- Stimulate reparative dentin formation
- Maintain pulp vitality
- Bactericidal/bacteriostatic
- Release fluoride to prevent secondary caries
- Adhere to dentin/restorative material
- Resist forces during restoration placement
- Provide appropriate seal
- Radiopaque

Technique

Tooth is isolated with rubber dam. Caries, if present, on side walls, are removed using spoon excavators. The exposed site is washed with weak disinfectant and covered with a moist cotton pellet dipped in disinfectant solution. Bleeding is controlled with 5.0% sodium hypochlorite. After controlling bleeding, pulp capping material is placed gently over the exposed site (Little pressure is used because displacement of capping material/dentin chips into the pulp may lead to further pulpal damage). The rest of the cavity is filled with interim restorative material (Fig. 30.3). It is established that bacterial leakage is responsible for pulpal response rather than toxicity of materials, which results in only mild and transitory pulpal response. Hypersensitivity to temperature change may persist for a day or so. Tooth is tested periodically for pulp vitality. Symptoms usually subside within 4–6 weeks. If tooth remains vital, a permanent restoration may be placed. If tooth becomes painful or becomes non-vital, root canal treatment becomes mandatory.

Materials Used

The materials used as pulp capping are:

a. Calcium hydroxide
b. Mineral trioxide aggregate (MTA)
c. Bonding agents
d. Resin-resorcinol
e. Antibiotics and corticosteroids
f. Growth factors
g. Enzymes
h. Stem cells
i. Propolis
j. Emdogain
k. TheraCal LC
l. Biodentine
m. Tricalcium phosphate
n. Miscellaneous

a. ***Calcium hydroxide:*** It is hypothesized that when vital pulp is covered with calcium hydroxide, it led to the formation of secondary dentin. Calcium hydroxide, when applied directly to pulp tissue, may lead to coagulation necrosis of adjacent tissues. Beneath the region of coagulation necrosis, cells of underlying pulp differentiate into odontoblasts, which help forming dentin matrix. However, a blood clot should not be left between calcium hydroxide and pulp. In such cases, hydroxyl ions trapped in the clot do not allow differentiation of odontoblasts. Calcium hydroxide maintains a local state of alkalinity necessary for bone/dentin formation. Earlier, it was postulated that calcium (Ca) from calcium hydroxide would diffuse into pulp and participate in the formation of reparative dentin. Later, it was established that calcium ions from calcium hydroxide do not enter into the formation of new dentin; however, it may come from blood. Calcium hydroxide-induced dentinal bridge contains multiple tunnel defects facilitating

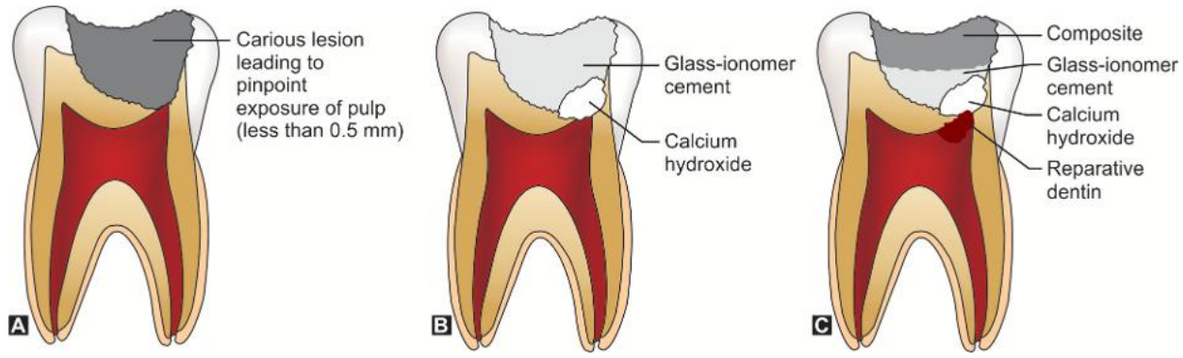


Fig. 30.3: Direct pulp capping: **(A)** Pinpoint carious exposure; **(B)** Placement of pulp capping material; **(C)** Formation of reparative dentin

bacterial microleakage. Also calcium hydroxide gets softened and allows leakage, resulting in pulpal inflammation and necrosis after one to two years. Such teeth may show evidence of calcification or internal resorption.

b. ***Mineral trioxide aggregate (MTA):*** Mineral trioxide aggregate (MTA) is of three types; viz. gray, white and modified. Any type can be used as pulp capping material.

MTA powder is mixed with sterile water in a ratio of 3:1 on a glass slab or paper pad. It is placed over exposure site using plastic/metal spatula. The chamber is filled with appropriate interim restorative materials. Dry MTA loses its cohesiveness and becomes hard to handle. When compared with calcium hydroxide, MTA produces effective dentinal bridge in a short period of time with significantly less inflammation. It resists microleakage better than calcium hydroxide. Presence of blood has little impact on leakage of MTA.

Properties

- Setting time: Sets within 24 hours
- Compressive strength: 70 MPa after one day
- pH of the set MTA: 11–13 after setting
- Biocompatible and non-mutagenic
- Antibacterial effect is similar to calcium hydroxide

c. ***Bonding agents:*** Adhesive bonding agents provide an effective biological seal against immediate and long-term microleakage required for pulp healing. However, when adhesive agents were compared with calcium hydroxide, they showed more pulp necrosis. Histological studies have shown persistent inflammatory reactions and hyaline alteration of extracellular matrix inhibiting bridge formation when bonding agents were used. Adhesive bonding agents are not preferred as pulp capping agent.

d. ***Resin-resorcinol:*** Resin-resorcinol has also been successfully tried as pulp capping material. The exposure site is washed immediately with normal saline solution and isolated with sterilized cotton rolls. The exposure site and the rest of the cavity is wiped with cresophene. Two drops of freshly prepared resinifying solution is poured over the exposure site and allowed to set. The rest of the cavity is filled with appropriate restorative material.

The effectiveness of resinifying agent as pulp capping material has been histologically proved. Histological sections revealed areas of fibrosis and hyalinization in the pulp, especially in the coronal area; however, in the middle and apical third of the sections, normal pulpal tissue was evident. The fibroblastic activities present at the coronal sites are suggestive of the healing process. The normal pulp tissue, without any sign of necrosis, present in the middle and apical part of the root confirms that the resinifying agent can successfully be used in direct pulp capping.

e. ***Antibiotic and corticosteroids:*** Antibiotics and corticosteroids in various combinations have also been used. These effectively provide bactericidal effect, which may lead to partial bridge formation. The narrowing of root canals has also been reported with their use. The material is clinically not preferred.

f. ***Growth factors:*** Growth factors regulate growth and development and also induce wound healing and tissue regeneration. Various forms of growth factors have been used as pulp capping agents to induce repair of pulp.

Bone morphogenic protein (BMP), belonging to β-transforming growth factor (TGF-β), is a potent modulator of tissue repair. Bone morphogenic proteins play a vital role in the differentiation of adult pulp cells into odontoblasts during pulpal healing.

The differentiated odontoblasts lead to deposition of osteodentin and tubular dentin. Other growth factors, such as insulin-like growth factor-I, epidermal growth factor, insulin-like growth factor II, platelet-derived growth factor have also been tried. Insulin- like growth factor-I has shown dentin bridge formation.

g. ***Enzymes:*** Enzymes, such as heme-oxygenase-1 and simvastatin, are being tried as pulp capping agent because of their potential to induce functional odontoblasts and improved dentin formation. Earlier, alkaline phosphatase when applied to exposed pulp stimulated the differentiation of other pulp cells into odontoblast, subsequently stimulating dentin matrix formation. It has been reported that heme oxygenase-1 (HO-1) might play a cytoprotective role against pro- inflammatory cytokines and nitric oxide in human pulp cells. Simvastatin improves the osteoblast function and suppresses osteoclast function, resulting in enhanced bone dentin formation. It stimulates the function of odontoblasts, thus leading to improved dentin formation. The enzymes induce angiogenesis and increase neuronal cells, indicating the possible effectiveness in pulp regeneration along with dentin regeneration. It has an anti-inflammatory effect; considered as an active pulp capping material to accelerate reparative dentin formation.

h. ***Stem cells:*** Stem cells are a group of undifferentiated biological cells, which are capable of self-renewal and multilineage differentiation. Their potential to stimulate tertiary dentin formation is effectively explored as pulp capping material.

Stem cells from pulp and human exfoliated deciduous teeth have shown efficacy of self-renewal and multi-lineage differentiation.

The disadvantage of stem cells might be their rejection because of immunological reasons (Stem cells are derived from embryos that are not patients own and the patient body may reject these cells).

i. ***Propolis:*** Propolis is a natural biological derivative containing flavonoids, phenolics, iron, zinc and other aromatic compounds. It is traditionally used as anti-inflammatory and antibacterial agent.

Propolis has great potential as pulp capping agent. It has been effectively used as an antioxidant, reducing hypersensitivity. Propolis has shown comparable results with MTA and Dycal, as regard dentin bridge formation. It may exhibit mild to moderate inflammation, which stimulates dentin bridge formation.

j. ***Emdogain:*** Emdogain is enamel matrix derivative secreted from Hertwig's epithelial root sheath during porcine tooth development. It is rich in amelogenin and amelin protein, capable of inducing reparative process. It stimulates the regeneration of acellular cementum, periodontal ligaments, and alveolar bone.

Emdogain contains biomolecular proteins, which promote odontoblast differentiation and reparative dentin formation. It has been established that Emdogain suppresses the inflammatory cytokine production and contains β-TGF growth factor; create a favorable environment for promoting wound healing in the pulp. Emdogain is used as an adjunct along with other pulp capping agents to achieve desired results.

Various investigators have reported that MTA produced a better quality reparative hard tissue with the adjunctive use of Emdogain as compared with calcium hydroxide.

Emdogenic gel when applied directly to exposed pulp without the use of other pulp capping agent could not produce the requisite hard tissue barrier (Emdogain possesses poor sealing ability).

k. ***TheraCal LC:*** TheraCal LC is a light-cured, resin-modified calcium silicate liner, mainly used in direct/indirect pulp capping, as a protective base/liner under restorative materials. Being radiopaque, the material can be easily identified during routine radiographic examination. TheraCal LC contains tricalcium silicate particles in a hydrophilic monomer, providing significant calcium release. Calcium release stimulates dentin bridge formation. The alkaline pH promotes apatite formation.

The barrier provides insulation and is moisture resistant.

It is established that TheraCal LC displayed higher calcium releasing ability and low solubility than MTA or Dycal. TheraCal LC can be cured to a depth of 2.0 mm, which stabilizes the material at the site of placement.

l. ***Biodentine:*** Biodentine is bioactive cement, simulating dentin in its physical properties. A modified form of calcium silicate cement, it is mainly used as dentin substitute reinforcing lost coronal and radicular dentin.

Biodentine has shown promising results as an alternative to calcium hydroxide. It is biocompatible with high strength; also exhibit excellent marginal adaptability. Its strong antibacterial activity provides effective results when placed over deep carious lesions.

It has also been used as retrograde filling materials, pulp capping agent, in internal/external resorption and also in apexification procedures.

It has been established that Biodentine induces secondary dentin formation by stimulating odontoblastic cell differentiation. Biocompatibility and capability of providing bacteria-free seal make Biodentine a material of choice in pulp capping/pulpotomy procedures.

Pulp capping: Key points

- Avoid exposing the pulp
- Even if residual caries present, cover carious dentin and seal properly
- Follow-up for at least six weeks for any symptoms
- If pulp gets exposed, control hemorrhage with appropriate astringent agents (sodium hypochlorite is effective hemostatic agent and is also antibacterial)
- Calcium hydroxide and mineral trioxide aggregate (MTA) are suitable pulp capping agents
- Newer agents, such as, morphogenic proteins, enzymes, plant extracts, resorcinol, etc. are also effective
- Zinc oxide eugenol, glass-ionomer cement and bonding agents are not successful (should be avoided)
- The cavity should be sealed immediately after pulp capping; protect underlying tissues from leakage and bacterial contamination
- The tooth should be assessed clinically and radiologically at periodic intervals; restore with suitable restorative material

Table 30.2: Pulp capping agents: Advantages and disadvantages

Pulp capping agent	*Advantages*	*Disadvantages*
Zinc oxide eugenol	• Reduces inflammation • Acts as obtundant	• Lack of calcific bridge formation • Eugenol release might be cytotoxic
Calcium hydroxide	• Accepted pulp capping material • Antibacterial (high alkalinity) • Induction of mineralization (calcific bridge formation) • Low cytotoxicity	• Highly soluble in oral fluids (potential of dissolution over time) • Lack of adhesion • Acid etching may degrade layer of calcium hydroxide
Mineral trioxide aggregate (MTA)	• Less pulpal inflammation • Hard tissue barrier formation is better than calcium hydroxide • Antibacterial • Radiopaque • Releases bioactive dentin matrix proteins • Biocompatible	• Poor handling characteristics • Long setting time • Gray MTA may cause tooth discoloration • High solubility
Bonding agents	• Provide adhesion to hard tissues • Provide effective seal against microleakage	• Absence of calcific bridge formation • May lead to chronic inflammatory pulpal response • Potential of toxicity
Resin-resorcinol	• Excellent sealing ability	• Calcific bridge formation is natural, may be due to bacteria-free seal
Polycarboxylate cement	• Chemically bond to tooth structure	• Does not stimulate calcific bridge formation
Isobutyl cyanoacrylate	• Reduces pulp inflammation	• Dentin bridge formation is not satisfactory
Corticosteroids and antibiotics	• Reduce pulp inflammation • Combination may lead to reparative dentin bridge	• Should not be used in patients at risk from bacteremia
Growth factors [insulin-like growth factor I, bone morphogenic protein (BMP)]	• Formation of osteodentin and tubular dentin • Superior to calcium hydroxide as regard mineralization inducing properties (formation of more homogeneous reparative dentin)	• Fail to stimulate reparative dentin in inflamed pulp • Technique sensitive (appropriate dose required for positive results) • Possibility of immunological problems • All growth factors may not induce reparative dentin formation
Enzymes (alkaline phosphatase, heme oxygenase 1, simvastatin)	• Anti-inflammatory action • Promote angiogenesis • Improve the function of odontoblasts; lead to superior dentin formation	• May lead to pulp tissue damage • Sensitive to appropriate concentration when applied directly to pulp tissue
Stem cells (pulp and other dental origins)	• Regeneration of dentin–pulp complex	• Technique sensitive • Still under investigations
Propolis	• Superior bridge formation (stimulates reparative dentin formation) • Reduces pulp inflammation and degeneration • Antioxidant, antibacterial, antifungal, antiviral and anti-inflammatory properties	• May exhibit mild/moderate inflammation after 2 to 4 weeks
Emdogain	• Induces odontoblast differentiation and reparative dentin formation • Suppresses the inflammatory cytokine production	• Emdogain gel when applied on exposed pulps without the adjunctive use of a pulp-capping material was ineffective in producing a hard tissue

Contd...

Table 30.2: Pulp capping agents: Advantages and disadvantages (*Contd...*)

Pulp capping agent	*Advantages*	*Disadvantages*
	• Creates a favorable environment for promoting pulp healing • Amount of hard tissue formed with Emdogain was twice as compared to calcium hydroxide	barrier (effective only as an adjunct)
TheraCal LC	• Stimulates regeneration of dentin–pulp complex • Bonds to deep moist dentin • Minimum solubility (solubility lower than MTA and Dycal) • Radiopaque • Displays higher calcium releasing ability	• Opaque and 'whitish' color may show through composite/ceramic materials affecting esthetics
Biodentine	• Biocompatible • Good antimicrobial activity • Stimulates tertiary dentin formation • Less soluble • Produces bacteria-free seal comparable to calcium hydroxide • Less setting time and good handling characteristics	• Long-term evaluation is not documented
Dentin shavings	• Induce secondary dentin deposition	• Placing and creating dentin shavings is difficult
Tricalcium phosphate	• Significant absence of pulp inflammation • Good physical properties • Induces calcific bridge formation	• Placement is difficult
Lasers	• Stimulate secondary dentin formation • Effective sterilization of targeted tissue • Bactericidal	• May cause thermal damage to pulp • Technique sensitive • Secondary dentin formation is natural under sterile environment
Miscellaneous (castor oil bean cement)	• Antibacterial • Less cytotoxic • Promote tissue healing • Induce calcific barrier	• Bioinert rather than bioactive • Technique sensitive

m. ***Tricalcium phosphate:*** Tricalcium phosphate, when implanted in bone, stimulates new bone formation. It does induce dentin bridge formation; however, severe initial inflammation may lead to necrosis of pulp. Practically, it has been replaced by MTA.

n. ***Miscellaneous:*** A few materials like Endosequence, castor oil bean cement, calcium enriched mixture (endodontic cement), freeze-dried bone matrix, etc. have also been tried as pulp capping materials, with varying success.

The advantages and disadvantages of pulp capping materials are summarized in Table 30.2.

3. Pulp Curettage

It is well documented that pulp exposure due to caries usually leads to pulp necrosis with passage of time. During caries removal, carious dentin chips may be inadvertently pushed into pulp tissue resulting in inflammation, resorption and encapsulation of dentin chips, which after capping may show foreign body reaction. It is suggested to remove superficial layer of pulp tissue by enlargement of the exposure site. This procedure is known as 'pulp curettage'. It is followed by control of hemorrhage, placement of a pulp capping material and sealing of the cavity as in direct pulp capping.

4. Partial Pulpotomy

It is the removal of only outer layer of hyperemic pulp tissues in exposed pulps. If 2.0–3.0 mm of pulp is removed, the procedure is called 'partial pulpotomy' (Fig. 30.4).

Indications

- Young tooth with a wide open apex and very thin root dentin walls.
- If sufficient tooth structure is present to allow proper restoration.

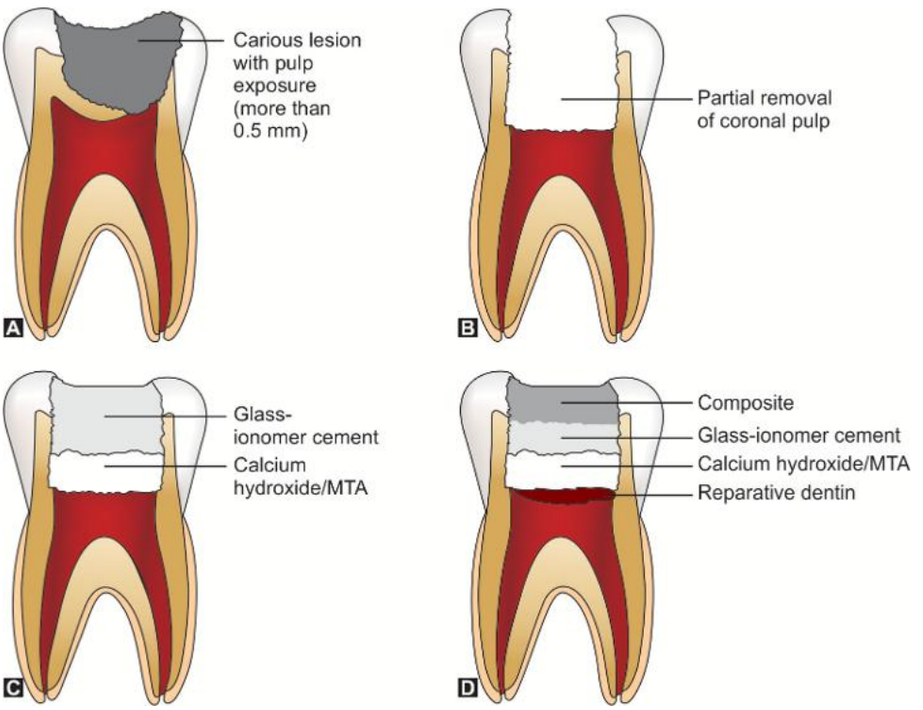


Fig. 30.4: Partial pulpotomy after carious exposure: **(A)** Pulp exposure; **(B)** Partial removal of coronal pulp; **(C)** Placement of calcium hydroxide/MTA; **(D)** Formation of reparative dentin

Contraindication

- If the exposure is very large or when more than two weeks have passed between injury and treatment time.

Advantages

- It preserves cell-zone due to minimal excision; hence, chances of better healing.
- Physiologic apposition of dentin is maintained.
- Natural color/translucency of tooth is maintained.
- Less chance of root canal obliteration.

Partial Pulpotomy for Carious Exposures

Procedure

- The inflamed pulp tissue beneath an exposure is removed to a depth of 1.0 to 3.0 mm or deeper to reach healthy pulp tissue.
- Pulpal bleeding is controlled by a bactericidal agent, such as sodium hypochlorite.
- Site is covered with calcium hydroxide or MTA.
- Calcium hydroxide has shown long-term success; MTA has shown more predictable dentin bridging and pulp health.

Partial Pulpotomy for Traumatic Exposures (Cvek Pulpotomy)

Procedure

- The procedure is similar to partial pulpotomy for carious exposures. In case of MTA, white, rather than gray, is recommended, especially in anterior teeth to decrease the chance of discoloration (Although both the versions have shown similar properties).
- Tooth is sealed with an appropriate restorative material to prevent microleakage.
- If hemostasis is not achieved, pulpotomy should be performed.

5. Pulpotomy

Pulpotomy procedure implies surgical removal of partly/completely inflamed coronal pulp and protecting the radicular pulp with a suitable material to retain its vitality. It is followed by filling the coronal chamber with a suitable base and restoring the tooth with a material that seals the tooth from microleakage. [For details, refer author's book, Essentials of Endodontics (2nd Edition): Vimal K Sikri]

Advantages

- Procedure is quick and easy to perform.
- Maintains the natural tooth color and preserves the tooth structure for better retention of restoration.
- Advantageous over complete pulpotomy in the preservation of cell-rich coronal pulp tissue.
- Successful outcome will allow the continuation of normal development of the tooth, including further root development and maturation.

PROGNOSIS

The outcome of the treatment of a deep carious lesion depends on various factors:

- Reparative capacity of pulp–dentin complex
- Soundness of the dentin in the preparation floor and walls
- Reparative capacity of affected dentin
- Extent of degeneration in the pulp–dentin complex
- Sealability of the restorative materials
- Potential of further irritation to the pulpodentinal complex resulting from instrumentation, restorative materials and procedures.

BIBLIOGRAPHY

1. Bjorndal L. Dentin and pulp reactions to caries and operative treatment: Biological variables affecting treatment outcome. Endodontic Topics 2002;2:10–19.
2. Bjorndal L. The caries process and its effect on the pulp: The science is changing and so is our understanding. J Endod 2008;34:52–58.
3. Bjorndal L. Dental caries: Progression and clinical management. Oper Dent 2002;27:211–16.
4. Chen L. Cytotoxicity and biocompatibility of resin-free and resin-modified direct pulp capping materials: a state-of-the-art review. Dent Mater J 2017;36:1–7.
5. da Rosa WLO, Cocco AR, da Silva TM, Mesquita LC, Galarca AD, da Silva AF, Piva E. Current trends and future perspectives of dental pulp capping materials: a systematic review. J Biomed Mater Res Part B 2017;1–11.
6. Derksen EK, Chen CF, Majewski R, Tootla RGH, Boynton JR. Reinforced Zinc Oxide Eugenol pulptomy: A retrospective study. Pediat Dent 2013;35:43–47.
7. Desouza CA, Hebling J, Hanks CT. Current status of pulp capping with dentin adhesive systems: a review. Dent Mater 2000;16:188–97.
8. Didilescu A, Cristache CM, Andrei M, Voicu G, Perlea P. The effect of dental pulp-capping materials on hard-tissue barrier formation. JADA 2018;1–17.
9. Fachin EVF, Filho EO, Cavarlho TP. Caries removal based on pulp biology. Electronic Journal of Endod Rosario 2007;1:1–5.
10. Farrugia C, Lung CYK, Wismayer PS, Arias-Moliz MT, Camilleri JC. The relationship of surface characteristics and antimicrobial performance of pulp capping materials. JOE 2018;1–6.
11. Hayashi M, Fujitani M, Yamaki C, Moncoi Y. Ways of enhancing pulp preservation by stepwise excavation-a systematic review. J Dent 2011;39:95–99.
12. Hilton TJ. Keys to clinical success with pulp capping: A review of literature. Oper Dent 2009;34:615–19.
13. Imazato S, Kitagawa H, Tsuboi R, Kitagawa R, Thongthai P, Sasaki J. Non-biodegradable polymer particles for drug delivery: A new technology for "bio-active" restorative materials. Dent Mater J 2017; 36:524–32.
14. Javaheri M, Kambakhsh S, Elemad-Moghadam S. Efficacy of two caries detector dyes in the diagnosis of dental caries. J Dent 2010;7:71–75.
15. Lee JH, Lee HH, Kim HW, Yu JW, Kim KN, Kim KM. Immunomodulatory/anti-inflammatory effect of ZOE-based dental materials. Dent Mater 2016;e1–e12.
16. Long Y, Liu S, Zhu L, Liang Q, Chen X, Dong Y. Evaluation of pulp response to novel bioactive glass pulp capping materials. J Endod 2017;1–4.
17. Marending M, Attin T, Zehnder M. Treatment options for permanent teeth with deep caries. Swiss Dent J 2016; 126:1007–18.
18. Medeiros E, Rosemblatt A. Differential diagnosis between active and arrested decayed lesion in dentin. Odontolgia 2006;5:27–31.
19. Murray P, Godoy F. The incidence of pulp healing defects with direct capping materials. Am J Dent 2006;19:171–76.
20. Negm AM, Hasanien EE, Abu-Seida AM, Nagy MM. Biological evaluation of a new pulp capping material developed from Portland cement. Experimental and Toxicologic Pathology 2016;1–8.
21. Poimenova A, Kitraki E, Kakaboura A, Rahiotis C. Early response of human pulp to direct capping with resin adhesive systems and calcium hydroxide. Dent Mater 2018;e1–e10.
22. Quershi A, Soujanya E, Nandkumar. Recent advances in pulp capping materials: An overview. J Clinical Diag Res 2014,8:316–21.
23. Ricketts D. Management of the deep carious lesion and the vital pulp dentin complex. Br Dent J 2001;191:606.
24. Ritter AV, Browning WD: Partial Caries Excavation. J Esthet rest dent 2012;24:148–52.
25. Rocas IN, Alves FRF, Rachid CTCC, Lima KC, Assuncao IV, Gomes PN, Siqueira JF. Microbiome of deep dentinal caries lesions in teeth with symptomatic irreversible pulpitis. PLOS ONE 2016;1–13.
26. Sikri V, Sikri P. Direct pulp capping with resorcinol: a clinical and histological evaluation. JCD 1998;1:90–94.
27. Thompson V, Craig R, Cuaro F, Green W, Ship J. Treatment of deep carious lesions by complete excavation or partial removal: A critical review. J Am Dent Assoc 2008;139:705–09.
28. Tomas-Catala CJ, Collado-Gonzalez M, Garcia-bernal D, Onate-Sanchez RE, Forner L, Llena C, Lozano A, Moraleda JM, Rodriguez-Lozano FJ. Biocompatibility of new pulp-capping materials NeoMTA plus, MTA repair HP, and Biodenine on human dental pulp stem cells. J Endod 2017;1–7.
29. Tran XV, Gorin C, Willig C, Baroukh B, Pellat B, Decup F, Vital SO, Chaussain C, Boukpessi T. Effect of a calcium silicate based restorative cement on pulp repair. J Dent Res 2012;91:1166–70.
30. Vural UK, Kiremitci A, Gokalp S. Randomized clinical trial to evaluate MTA indirect pulp capping in deep caries lesions after 24-months. Oper Dent 2017;42:470–77.